

AXONA

API Reference

Every public symbol exported from @axona/protocol v2.11.0

David A. Smith
Axona.net

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Reference for every public symbol exported from @axona/protocol v2.11.0. Organized by what application developers actually reach for; deep-customization surfaces are at the end.

Companion documents:

- [Quick Start](#) — 5-minute working roundtrip.
- [Programmer Guide](#) — mental model + worked example + pitfalls.
- [Security changelog](#) — what each kernel version protects (authenticated handshake, channel binding, pub/sub trust boundary, verified routing admission). See §17 below for the developer-facing summary of the security model.

Imports throughout assume:

```
import { /* ... */ } from '@axona/protocol';
```

Sub-path imports (e.g. @axona/protocol/contracts/Transport.js) work for symbols that need it but most apps only need the main barrel.

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Types referenced throughout

[Identity](#identity-type), [Envelope](#envelope-type), [Subscription](#subscription-type), [Peer](#peer-id-type), [TopicId](#topic-id-type).

Application surface

Types

Identity (type)

```
{
  id:      string;           // 66-char lowercase hex (264-bit nodeId
                             )
  pubkey:   Uint8Array;      // 32-byte Ed25519 public key
  pubkeyHex: string;        // 64-char lowercase hex
  privateKey: CryptoKey;    // Web Crypto Ed25519 (browser + Node
                             20+)
  region:   { lat: number, lng: number };
  createdAt: number;        // ms epoch
}
```

The top 8 bits of `id` are `geoCellId(lat, lng, 8)` (the S2 cell at level 8); the bottom 256 bits are `sha256(pubkey)`.

Envelope (type)

```
{
  msgId:      string;           // 64-char hex sha256 of canonical
    inputs
  seq:        number;           // per-publisher monotonic sequence (
    kernel >= v2.9.0)
  ts:         number;           // ms epoch when published
  topic:      string;           // the application-level topic name
  message:    any;              // any JSON-serializable payload
  signerPubkey?: string;        // 64-char hex; present iff signed
  signature?: string;           // "ed25519:<128 hex>"; present iff
    signed
}
```

`seq` (envelope format **v2**, kernel $\geq$ v2.9.0) is folded under the signature alongside `ts`. The kernel uses it for **replay/freshness** (a captured envelope can't be re-injected as live once its signed `ts` ages out of the ~5-min window) and for **deterministic ordering** ("latest", bounded-queue eviction). You don't set it — `peer.pub` stamps it.

A **retraction** (see `peer.kill`) is delivered to your `sub` handler as a minimal marker instead of a full envelope:

```
{ deleted: true, msgId: string, topic: string | null }
```

Branch on `env.deleted === true` to drop your local copy of `msgId`.

Subscription (type)

Opaque handle returned by `peer.sub()`. Has:

```
sub.id          // string --- unique per subscription
sub.topicId     // string --- 66-char hex topic ID
sub.topicName   // string --- your original topic name
sub.stop()      // Promise<void> --- cancel; idempotent
```

PeerId (type)

In the kernel's public API, peer IDs are **66-char lowercase hex strings**. The synaptome stores them as `bigint` internally; conversion happens at the API boundary. Pass strings unless a function's signature specifically says `BigInt` (`peer.lookup`, the synapse admit example).

TopicId (type)

Same shape as `PeerId` — 66-char lowercase hex. Top 8 bits encode the addressing-mode S2 prefix, bottom 256 bits are sha256 of the canonical input.

1. Identity

`deriveIdentity({ lat, lng, extractable? }) → Promise<Identity>`

Generate a fresh 264-bit Ed25519 identity. Top 8 bits of the resulting `id` encode the S2 cell containing (`lat`, `lng`); the bottom 256 bits are SHA-256(pubkey), so the `nodeId` is **self-authenticating** — a peer can only claim an `id` it holds the private key for (this is what the axona/4 handshake checks; see §17).

```
const id = await deriveIdentity({ lat: 38.0, lng: -77.0 });
console.log(id.id);           // 'df7c5e...' (66 hex chars, us-east
    prefix)
console.log(id.id.slice(0, 2)); // 'df'      (us-east S2)
```

****Optional extractable**** (default `true`): pass `false` to hold the signing key as a **non-extractable** `CryptoKey` — it can sign but cannot be read back out by page script, so an XSS incident or a malicious dependency cannot exfiltrate the long-term identity. Recommended for ephemeral browser identities. Leave `true` only when you need to `dumpIdentity` it for persistence.

Returned `Identity`: { `id` (66-hex), `pubkey` (`Uint8Array`), `pubkeyHex` (64-hex), `privateKey` (`CryptoKey`), `region`, `createdAt`, `sign(bytes)`, `verify(bytes, sig)` }.

Throws `IdentityError` if Web Crypto isn't available or `lat/lng` are out of range.

`dumpIdentity(identity) → Promise<envelope>`

Serialize an `Identity` to a JSON-safe envelope (the `privateKey` is exported as base64 PKCS#8).

```
const env = await dumpIdentity(identity);
// {
//   id:      '<66 hex>',
//   pubkey:  '<64 hex>',
//   privkey: '<base64 PKCS#8 string>',
//   region:  { lat, lng },
//   createdAt: <ms>,
// }
localStorage.setItem('my-identity', JSON.stringify(env));
```

`loadIdentity(envelope) → Promise<Identity>`

Inverse of `dumpIdentity`. Reconstructs the full `Identity` from the JSON envelope by importing the PKCS#8 key.

```
const env = JSON.parse(localStorage.getItem('my-identity'));
const identity = await loadIdentity(env);
```

Throws `IdentityError` (`IDENTITY_LOAD_FAILED`) if the envelope is malformed or the `pubkey` doesn't match the `id`'s hash.

2. *AxonaPeer construction + lifecycle*

The peer is the per-node DHT contract implementation.

`new AxonaPeer({ domain, node, identity, transport?, axonaManager?, persist?, engine? })`

Construct a peer. Required:

- `domain` — an `AxonaDomain` instance (shared routing parameters).
- `node` — a `NeuronNode` (or any object with {`id`, `alive`, `transport`, `synaptome`}).

- `identity` — from `deriveIdentity`; needed for signed publishes.

Optional:

- `transport` — overrides `node.transport`. The peer uses it directly for `send/notify/lookup` round-trips.
- `axonaManager` — pre-built `AxonaManager`. If omitted, the peer resolves one on first `pub/sub` via `engine.axonaManagerFor(node)`.
- `persist` — a `PersistenceAdapter`; enables auto-checkpointing of identity envelope, subscriptions, and synaptome snapshots.
- `engine` — legacy; pass `null` for new code.

```
const peer = new AxonaPeer({
  domain:    new AxonaDomain({ k: 20 }),
  node:      new NeuronNode({ id: BigInt('0x' + id.id), lat, lng }),
  identity:  id,
  transport,
});
```

`peer.start()` → `Promise<void>`

Bring the peer up — installs wire-level routing handlers, sets up the delivery hook, and (if `persistence` is wired) hydrates any persisted state.

Idempotent. Returns the peer once started.

`peer.stop()` → `Promise<void>`

Tear down. Drains in-flight `pub` calls, stops timers, releases listeners. Idempotent.

`peer.join(opts?)` → `Promise<void>`

Production bootstrap: discover the network, populate synaptome to target size, ready for routing. Optional `opts` customize fan-out parameters; defaults are usually fine.

In simple in-process setups (where you admit synapses directly), you don't need `join()` — `start()` is enough.

`peer.leave({ drain?, notify?, timeoutMs? })` → `Promise<void>`

Graceful shutdown. With `drain: true` (default), waits up to `timeoutMs` (default 5000) for in-flight `pubs` to complete. With `notify: true` (default), sends `pubsub:unsubscribe-k` for every active subscription so axons can sweep our entries promptly.

```
await peer.leave({ drain: true, notify: true, timeoutMs: 3000 });
```

`peer.getNodeId()` → `string` | `bigint`

Returns whatever was passed as `node.id` at construction time. If you followed the convention of passing a `BigInt` to `NeuronNode`, this returns `BigInt`; if you passed a `string`, it returns a `string`.

3. Pub/sub

`peer.pub(topicName, message, opts?)` → `Promise<msgId>`

Publish a message to a topic.

Arg	Type	Notes
<code>topicName</code>	<code>string</code>	Application-defined name (gets hashed).
<code>message</code>	<code>any</code>	JSON-serializable. Strings, numbers, objects, arrays all OK. Size: see below.
<code>opts.publisher</code>	<code>string</code> <code>null</code> <code>undefined</code>	Addressing mode. See §6.
<code>opts.sign</code>	<code>boolean</code>	Default <code>true</code> . Set <code>false</code> for anonymous publishes (no <code>signerPubkey/signature</code> in envelope).

Returns: 64-char hex `msgId` — sha256 of the canonical envelope inputs. Subscribers see this same id in `env.msgId`.

Maximum message size: 256 KiB (kernel $\geq$ v2.8.1; was 64 KiB before). The limit is on the *serialized envelope* (`JSON.stringify({ msgId, ts, topic, message, signature, signerPubkey })`), so your usable payload is 256 KiB minus a few hundred bytes of envelope overhead. It's measured on string length, so heavily multi-byte (CJK/emoji) content can exceed 256 KiB on the wire.

peer.pub rejects an oversized message up front (kernel $\geq$ v2.8.2) with `PublishError(PUBLISH_PAYLOAD_TOO_LARGE)` — you get an immediate, catchable error rather than a message that silently fails to arrive. The root axon also enforces the same cap at ingress as defense-in-depth.

Pub/sub is a *broadcast* path (replicated to K root axons, cached, fanned to all subscribers), so it is the wrong place for images/documents even under 256 KiB. For binary content, publish a small **content-reference** (hash + size + mime) and transfer the bytes out-of-band, content-addressed.

```
const synth = '<region S2 hex>' + '0'.repeat(64);
const msgId = await peer.pub('news/world', { title: 'hi' }, {
  publisher: synth,
});
```

Throws `PublishError`:

- `PUBLISH_INVALID_TOPIC` — empty topic name.
- `PUBLISH_SIGN_FAILED` — signing failed (identity missing privateKey, Web Crypto unavailable, etc).
- `PUBLISH_PAYLOAD_TOO_LARGE` — serialized envelope exceeds the 256 KiB cap (kernel $\geq$ v2.8.2; see "Maximum message size" above).
- `PUBLISH_INVALID_MESSAGE` — the message is **not JSON-serializable** (circular reference, `BigInt`, etc.), so the envelope can't be encoded.

`peer.sub(topicName, handler, opts?)` $\rightarrow$ `Promise<Subscription>`

Subscribe to a topic. `handler` is invoked for every delivered envelope.

Arg	Type	Notes
<code>topicName</code>	<code>string</code>	Must match what publishers use.
<code>handler</code>	<code>(envelope) => void</code>	Called with the full <code>Envelope</code> .
<code>opts.publisher</code>	<code>string \{\} null \{\} undefined</code>	Must match publisher's mode.
<code>opts.since</code>	<code>'all' \{\} 'latest' \{\} number \{\} undefined</code>	Replay mode. See below.

****since modes**:**

since	What you get
omitted / <code>undefined</code>	Live tail only — no cached messages.
<code>'all'</code>	Every message in the axon's replay cache, then live tail.
<code>'latest'</code>	The most recent ~1s of cache, then live tail.
<code><number></code>	Messages with <code>publishTs > since</code> (ms epoch).

Returns: a `Subscription` handle. Call `.stop()` to cancel.

```
const sub = await peer.sub('news/world', (env) => {
  console.log(env.ts, env.message);
}, { publisher: synth, since: 'all' });

// later...
await sub.stop();
```

Your handler also receives **retraction markers** — `{ deleted: true, msgId, topic }` — when a publisher kills a message (see `peer.kill()`). Branch on `env.deleted === true` to drop your local copy:

```
const sub = await peer.sub('news/world', (env) => {
  if (env.deleted) { dropFromUI(env.msgId); return; }
  render(env.message);
}, { publisher: synth, since: 'all' });
```

Throws `SubscribeError`:

- `SUBSCRIBE_INVALID_TOPIC`
- `SUBSCRIBE_HANDLER_MISSING`

`peer.unsub(topicName, opts?)` → `Promise<{ ok, removed }>`

Unsubscribe from a topic by name (kernel $\geq$ v2.10.0) — the counterpart to `peer.sub`. Stops **every** local subscription you hold for the topic (no need to keep the `Subscription` handle) and, once the last one goes, sends the network unsubscribe so the topic's root axons drop you. Idempotent: unsubscribing a topic you're not on returns `{ ok: true, removed: 0 }`.

Arg	Notes
<code>topicName</code>	The topic to leave.
<code>opts.publisher</code>	Must match what you passed to <code>sub</code> (default = your own feed, <code>null</code> = public, <code>hex</code> = another publisher's feed).

```
await peer.unsub('news/world', { publisher: synth });
```

Self-only by construction — it removes only *your* `subscriberId` (the network enforces this), so it can't be used to unsubscribe anyone else. Throws `SubscribeError(SUBSCRIBE_INVALID_TOPIC)` on an empty topic.

`peer.pull(msgId, opts)` → `Promise<Envelope | null>`

Fetch an envelope from the K-closest axons. Returns `null` if it's not in any reachable cache window (which includes a message that has **expired** past its hold time — see [Topic limits](#)).

Arg	Notes
<code>msgId</code>	64-char hex from <code>pub</code> / a previous delivery — **or null/omitted to fetch the topic's most-recent message** (kernel $\geq$ v2.10.0).
<code>opts.topic</code>	The topic name.
<code>opts.publisher</code>	Same addressing mode used to publish.
<code>opts.timeoutMs</code>	Default 1000.

```
// a specific message...
const env = await peer.pull(msgId, { topic: 'news/world', publisher:
  synth });
// ...or the latest on the topic (no msgId):
const latest = await peer.pull(null, { topic: 'news/world', publisher:
  synth });
```

"Latest" is the highest-ordered message by signed `seq` (then `ts`, `msgId`). A successful pull also **slides the message's hold time** forward (now + hold), bounded by its absolute 48 h ceiling — reads keep a hot message alive, but can't pin it forever.

Throws `PullError` on transport failure (not on cache miss — that returns `null`). Passing a non-null, non-64-hex `msgId` throws `PULL_INVALID_MSGID`.

`peer.metrics(topicName, opts?) → Promise<metricsObj>`

Best-effort delivery, retention, and subscriber counts for a topic.

```
const m = await peer.metrics('news/world', {
  publisher: synth, // null for a public topic
  timeoutMs: 500,
});
// {
//   publishes:    <distinct messages ever seen>,
//   current_count: <events live in the tree right now>, // kernel >=
//                 v2.11.0
//   subscribers:  <max direct subscribers at a relay>, // live,
//                 kernel >= v2.11.0
//   deliveries:   <total fan-out deliveries>,
//   pulls:        <total pull() hits>,
//   reshares:     <total reshare bumps>,
//   relayCount:   <distinct relays that replied>,
// }
```

`current_count` is the number of published events **currently retained** (live — non-expired, non-killed) across the topic's tree, so it falls as messages are killed or age out under the hold-time TTL; `publishes` is the cumulative count of distinct messages ever counted, so it only rises. `subscribers` is the max direct-subscriber count reported by any single responding relay — exact for an unsplit topic, a lower bound once the tree splits into sub-axons. Values are aggregated from whichever axons reply within `timeoutMs`. Use for UX ("12 people in this room"), not for billing.

`peer.kill(topicName, msgId, opts?) → Promise<{ ok }>`

Retract a message you published (kernel $\geq$ v2.10.0) — "unsend". The topic's root axons accept the kill **only if it's signed by the same key that signed the original message** (creator-only, self-authenticating), drop it from their replay caches, tombstone the `msgId` so a lagging replica can't resurrect it, and forward a delete marker to current subscribers (which arrives on their `sub` handler as `{ deleted: true, msgId, topic }`).

```
const msgId = await peer.pub('news/world', { title: 'oops' }, {
  publisher: synth });
await peer.kill('news/world', msgId, { publisher: synth });
```

Arg	Notes
topicName	The topic you published to.
msgId	Required 64-char hex (the value <code>pub</code> returned). No "kill latest".
opts.publisher	Must match what you passed to <code>pub</code> .

! Best-effort redaction, **not** a cryptographic un-send. A subscriber who already received the message can keep it; an offline subscriber may never see the purge. And an anonymous (`sign: false`) message has no provable creator and so **cannot** be killed.

Throws `KillError`: `KILL_INVALID_TOPIC`, `KILL_INVALID_MSGID`, `KILL_SIGN_FAILED` (no identity — kills must be signed).

`peer.unpub(topicName, opts?)` → `Promise<{ ok }>`

Remove a topic's whole message queue (kernel $\geq$ v2.10.0) — **owner-only**. The owner is the identity whose `nodeId` seeds the topic id; root axons verify ownership self-authenticatingly (the signer's pubkey must bind to the owner `nodeId`, and that `nodeId` must derive the topic id).

Arg	Notes
opts.destroy	<code>false</code> (default) drops the messages, keeps the topic config; <code>true</code> is total removal (messages + config/ACL + ownership state).
opts.publisher	Owner selector; default = this peer.

```
await peer.unpub('news/world');           // clear the queue
await peer.unpub('news/world', { destroy: true }); // remove the topic
entirely
```

Public (ownerless) topics have no owner to prove ownership and **cannot** be unpublished. **Throws** `UnpubError`: `UNPUB_INVALID_TOPIC`, `UNPUB_PUBLIC_TOPIC`, `UNPUB_SIGN_FAILED`.

Topic limits — queue size & hold time

Every topic's replay queue is bounded (kernel $\geq$ v2.10.0):

- **Max messages** — default **100**, max **256**. When full, the lowest-ordered message (by signed `seq`) is evicted — deterministic, so all replicas converge. A max of **1** is a *retained* / *latest-value* slot (each publish replaces the prior one); pair it with `peer.pull(null, ...)`.
- **Hold time** — default **24 h**, hard ceiling **48 h**. A message past its hold expires and is swept; it stops being delivered or pulled. A `pull` slides the hold forward, bounded by the ceiling.

- **Per-publisher quota (open topics only)** — on a public/anyone-may-publish topic, one publisher can occupy at most $\frac{1}{4}$ of the queue, so a single flooder can't evict everyone else. Owner-gated topics aren't quota'd.

In Phase A these are protocol defaults; owner-set per-topic values arrive with the topic-config object in a later phase.

4. Direct messaging

For 1:1 messages between specific peers, bypassing pub/sub.

peer.send(peerIdHex, message) → Promise<reply>

Request/response. Resolves to whatever the recipient's `onMessage` handler returns (or rejects with the remote handler's thrown error).

```
const reply = await alice.send(bobIdHex, {
  kind: 'ping', body: 'hello',
});
// reply = whatever bob's handler returned
```

Throws `TypeError` if `peerIdHex` isn't 66-char hex.

peer.notify(peerIdHex, message) → Promise<boolean>

Fire-and-forget. Resolves once the frame is enqueued (NOT when delivered). Return value is `true` on successful enqueue, `false` if the transport rejected it.

```
await alice.notify(bobIdHex, { kind: 'typing', user: 'alice' });
```

peer.onMessage(handler) → void

Register the single direct-message handler. Calling `onMessage` twice **replaces** the previous handler — for multi-kind dispatch, do it inside the handler:

```
peer.onMessage(async (senderIdHex, message) => {
  switch (message?.kind) {
    case 'ping':
      return { reply: 'pong' }; // send() resolves to this
    case 'typing':
      console.log(`${senderIdHex} typing`);
      return; // notify() discards
  }
});
```

`senderIdHex` is set by the transport at the receiving end. It's the authenticated sender (per the transport's binding); for higher assurance verify a signed envelope inside `message`.

5. Mesh introspection

peer.peers() → string[]

Current synaptome membership as 66-char hex strings.

peer.onPeerJoin(handler) → () => void

Fires when a peer is admitted to the synaptome. Handler receives (peerIdHex, ctx). Returns an unsubscribe function.

```
const unsub = peer.onPeerJoin((peerId, ctx) => {
  console.log('admitted', peerId, 'via', ctx?.addedBy);
});
// later: unsub();
```

peer.onPeerLeave(handler) → () => void

Symmetric. Fires when a peer is evicted (TTL, churn, explicit unbind). Same signature.

peer.lookup(targetKey) → Promise<lookupResult>

Iterative XOR-routed walk to find targetKey (typically a peer ID, but any 264-bit key works).

```
const r = await peer.lookup(BigInt('0x' + targetHex));
// {
//   found: true,
//   hops: 3,
//   time: 47,    // ms; live RTT since v1.1.2
//   path: [<bigint>, <bigint>, ...], // the hop sequence
// }
```

targetKey may be a BigInt or a 66-char hex string — topicToBigInt converts internally.

peer.health() → healthObj

A cheap, synchronous diagnostic snapshot — safe to poll on a UI tick or from a status endpoint:

```
peer.health();
// {
//   nodeId:      '<66 hex>',
//   synaptomeSize: 6,           // routing-table entries
//   peers:       ['<66 hex>', ...], // same, as a list
//   subscriptions: 2,
//   axonRoles:   [{ topic, isRoot, children, cacheSize }],
//   wireVersion: '1.0',
//   started:     true,
//   transport: {                // null on sim/node; populated
//     on web
//   },
//   boundCount:  9,             // peers authenticated via axona
// }
```

```
//   meshChannels: 8,           // WebRTC data channels (any
//   state)
//   meshOpen:      8,           // open data channels
//   meshBound:     8,           // open AND axona/4-
//   authenticated
//   bridgeState:   'open',
// },
// meshDegraded: false,         // true => channels open but NOT
//   bound
// }
```

`meshDegraded` is the **routing-truth** signal: `true` means data channels are open but the `axona/4` handshake hasn't authenticated them, so no verified routing is flowing. A single `true` tick can be a normal mid-handshake transient; treat a value that stays `true` across several polls as the real signal. (`boundCount/meshBound` are the honest "usable peers" counts — distinct from the raw channel count.)

`peer.onLog(level, handler) → () => void`

Subscribe to structured log events at a given level. **The level comes first**; returns an unsubscribe function.

```
const off = peer.onLog('warn', (msg, ctx) => console.warn(msg, ctx));
// level in 'debug' | 'info' | 'warn' | 'error'
```

`peer.onError(handler) / peer.onUpgradeRequired(handler) → () => void`

`onError` fires on background `AxonaError` emissions; `onUpgradeRequired` fires (and also fan-routes through `onError`) on a wire-version mismatch, carrying the `UpgradeRequiredError` with `{ reason, serverVersion, clientVersion, downloadUrl }`.

```
peer.onError((err) => {});           // err = AxonaError
peer.onUpgradeRequired((err) => showUpgradeBanner(err.context.
  downloadUrl));
```

6. Topic IDs

Topic IDs share the same 264-bit address space as node IDs. The top 8 bits encode an S2 region; the bottom 256 bits are `sha256(canonical input)`.

`deriveTopicId(publisherId, topicName) → Promise<string>`

Compute the 66-char hex topic ID.

The kernel **doesn't validate** that a 66-char synthetic ID corresponds to a real peer — it's just used as the hash prefix. Use this to bind topics to a region without tying them to a specific publisher.

publisherId	Mode	Top 8 bits	Hash input
null or undefined or "<66 hex>" (real nodeId)	Public Publisher-keyed	00 (global bucket) publisher's S2 prefix	topicName publisherId + ':' + topicName
'<s2 hex><64 zeros>'	Region-keyed	region's S2 prefix	synth + ':' + topicName

```
import { deriveTopicId, geoCellId } from '@axona/protocol';

// Public mode
const pub = await deriveTopicId(null, 'world-news');
// '00' + sha256('world-news')

// Publisher-keyed
const pk = await deriveTopicId(myIdHex, 'posts');
// myS2 + sha256(myIdHex + ':posts')

// Region-keyed (recommended for most chat / forum / feed apps)
const s2 = geoCellId(lat, lng, 8);
const synth = s2.toString(16).padStart(2, '0') + '0'.repeat(64);
const rk = await deriveTopicId(synth, 'world-news');
// regionS2 + sha256(synth + ':world-news')
```

Throws `TypeError` if `topicName` is empty or `publisherId` is non-null and not 66-char hex.

Other pubsub/post.js exports

```
makePost({ topic, content, references?, identity?, sign? }) // -> SignedPost
verifyPostHash(post) // -> boolean
verifyTopicOwnership(post) // -> boolean
verifySignature(post) // -> Promise<boolean>
canonical(post) // -> string (JSON-canonicalized)
sha256Hex(str) // -> Promise<string>
```

These are lower-level helpers for protocol implementors building on top of the post-layer rather than `peer.pub`. Most apps don't need them.

7. Envelopes + verification

```
buildEnvelope({ topic, message, ts?, seq?, identity, sign?
}) → Promise<Envelope>
```

Construct an envelope explicitly. Useful when you want to sign a DM payload before sending it via `peer.send`, or to test verification flows.

```
const env = await buildEnvelope({
  topic:    'dm:to-bob',
  message:  { text: 'hi' },
  seq:      Date.now(),      // per-publisher monotonic; 0 if omitted
  identity, sign: true,
});
// env = { msgId, seq, ts, topic, message, signerPubkey, signature }
```

`seq` (envelope format v2) is folded under the signature. `peer.pub` supplies a monotonic value automatically; you only pass it when building envelopes by hand. The signature covers a **domain-tagged** core (`axona:pubsub-envelope:v2`) so it can't be replayed as another kind of signature.

Throws `TypeError` if `identity` is missing `privateKey` or `pubkeyHex` when `sign: true`.

```
verifyEnvelope(envelope) → Promise<{ ok, reason?, signed }>
```

Verify that `envelope.signature` is valid for the domain-tagged (`seq`, `ts`, `topic`, `message`) canonical inputs against `envelope.signerPubkey`, and that `msgId` matches. Requires `seq` (envelope format v2) — a pre-v2 envelope fails with `missing_seq`.

Returns a result object, not a boolean — check `.ok`:

```
const res = await verifyEnvelope(env);
if (!res.ok) console.warn('forged/invalid envelope', env.msgId, res.
  reason);
// res.signed === false for an unsigned envelope that is otherwise
  valid
// (res.ok === true). Your application decides whether to accept
  unsigned.
```

! A common mistake: if `!(await verifyEnvelope(env))` is **always false** — the return value is an object (truthy). Always test `.ok`.

`reason` (when `!ok`) is one of `not_an_object` / `missing_*` (incl. `missing_seq`) / `unknown_signature_scheme` / `wrong_key_or_signature_length` / `bad_signature` / `bad_msgid`. As of kernel v2.7.0, root axons also verify the publisher signature at ingress before fan-out (finding B-4); as of v2.9.0 they additionally reject stale replays (freshness window + per-publisher `seq`, finding C-2) — but apps should still verify what they consume.

`verifyEnvelope` checks the signature and `msgId`; it does **not** check

freshness (that's intentional — the legitimate replay-to-late-subscribers path serves cached history). Freshness is enforced only at live-publish ingress inside the kernel; the `checkFreshness(envelope, { now?, maxSkewMs? })` helper is exported if you need it.

`computeMsgId({ seq, topic, ts, message, signature? }) → Promise<string>`

Stand-alone `msgId` computation matching what `buildEnvelope` does (the canonical hash now binds `seq`). Useful for verifying a `msgId` server-side.

```
const msgId = await computeMsgId({
  seq: env.seq, topic: env.topic, ts: env.ts, message: env.message,
  signature: env.signature, // include for a signed envelope
});
console.log(msgId === env.msgId); // true
```

8. Errors

All thrown errors inherit from `AxonaError`:

```
class AxonaError extends Error {
  code: string; // stable UPPER_SNAKE identifier
  cause?: Error; // wrapped underlying error
  context?: object; // machine-readable details
}
```

Subclasses:

Class	Example codes
<code>IdentityError</code>	<code>IDENTITY_INVALID</code> , <code>IDENTITY_LOAD_FAILED</code> , <code>IDENTITY_KEY_IMPORT_FAILED</code>
<code>TransportError</code>	<code>TRANSPORT_NOT_STARTED</code> , <code>TRANSPORT_TIMEOUT</code> , <code>TRANSPORT_NOT_BOUND</code> , <code>TRANSPORT_CLOSED</code>
<code>PublishError</code>	<code>PUBLISH_INVALID_TOPIC</code> , <code>PUBLISH_SIGN_FAILED</code> , <code>PUBLISH_PAYLOAD_TOO_LARGE</code> , <code>PUBLISH_INVALID_MESSAGE</code>
<code>SubscribeError</code>	<code>SUBSCRIBE_INVALID_TOPIC</code> , <code>SUBSCRIBE_HANDLER_MISSING</code>
<code>KillError</code>	<code>KILL_INVALID_TOPIC</code> , <code>KILL_INVALID_MSGID</code> , <code>KILL_SIGN_FAILED</code>
<code>UnpubError</code>	<code>UNPUB_INVALID_TOPIC</code> , <code>UNPUB_PUBLIC_TOPIC</code> , <code>UNPUB_SIGN_FAILED</code>
<code>PullError</code>	<code>PULL_INVALID_MSGID</code> , <code>PULL_AXONS_UNREACHABLE</code>
<code>MetricsError</code>	<code>METRICS_AXONS_UNREACHABLE</code>
<code>UpgradeRequiredError</code>	<code>UPGRADE_REQUIRED</code> (bridge close code 4426)

The full code list is in `ErrorCodes`:

```
import { ErrorCodes } from '@axona/protocol';
```

```
console.log(ErrorCodes.PUBLISH_SIGN_FAILED); // 'PUBLISH_SIGN_FAILED'
```

Switch on `.code`, not `.message`:

```
try { await peer.pub(topic, msg); }
catch (err) {
  if (err.code === ErrorCodes.PUBLISH_SIGN_FAILED) {
    await reload();
  } else { throw err; }
}
```

Wire-format helpers

```
isWireError(obj)          // -> boolean --- does this look like an over-
  the-wire AxonaError?
fromWire(obj)             // -> AxonaError --- reconstruct
err.toWire()              // -> { __axonaError: true, class, code,
  message, context }
```

Errors thrown inside remote handlers (i.e., inside another peer's `onMessage` reached via `peer.send`) survive serialization with their class and code intact.

9. Persistence

PersistenceAdapter (interface)

```
class PersistenceAdapter {
  async load(key)          { /* returns previously-saved string or null
    */ }
  async save(key, value) { /* persists; value is string */ }
  async clear()            { /* wipe everything; idempotent */ }
}
```

Implement this to plug in your own storage (Redis, S3, postgres, etc.). Two built-in implementations:

createIndexedDbPersistence({ dbName }) → **PersistenceAdapter**

Browser. Uses a single object store inside the named DB.

```
import { createIndexedDbPersistence } from '@axona/protocol';

const peer = new AxonaPeer({
  domain, node, identity,
  persistence: createIndexedDbPersistence({ dbName: 'my-app' }),
});
```

`createFilePersistence({ path }) → PersistenceAdapter`

Node. Single JSON file on disk.

```
import { createFilePersistence } from '@axona/protocol';

const peer = new AxonaPeer({
  domain, node, identity,
  persistence: createFilePersistence({ path: './state.json' }),
});
```

Manual snapshots

```
const state = await peer.snapshot();
// state = { identity?, subscriptions, synaptome, lookups, ... }
// --- any JSON-safe object

await peer.fromSnapshot(state);

// static factory:
const peer = await AxonaPeer.fromSnapshot(state, { engine, node,
  transport });
```

Transport + protocol surface

10. Contracts

Abstract base classes that transports / DHT implementations extend. Application developers rarely use these directly — they're for authoring custom transports or alternate DHT engines.

Transport (abstract class)

```
import { Transport } from '@axona/protocol';

class MyTransport extends Transport {
  async start(localNodeId) { /* ... */ }
  async stop() { /* ... */ }
  async send(peerId, type, body) { /* ... */ } // request/response
  async notify(peerId, type, body) { /* ... */ } // fire-and-forget
  async openConnection(peerId) { /* ... */ }
  async closeConnection(peerId) { /* ... */ }
  isConnected(peerId) { /* ... */ }
  getLatency(peerId) { /* RTT ms or -1 */ }
  onRequest(type, handler) { /* ... */ }
  onNotification(type, handler) { /* ... */ }
  onPeerDied(handler) { /* ... */ }
}
```

The full contract surface is in `@axona/protocol/contracts/Transport.js` (sub-path import).

DHT (abstract class)

Used by AxonaManager's dht adapter. Defines `getSelfId`, `findKClosest`, `routeMessage`, `sendDirect`, `onRoutedMessage`, `onDirectMessage`. See `axona-peer/src/browser_engine.js` for a concrete adapter that wraps `AxonaPeer`'s methods.

BootstrapService (abstract class)

For implementing peer discovery (signaling brokers, seed lists, DNS-SD, etc). Rarely needed by app code.

11. Sim transport

In-process router for testing + simulators.

`new SimNetwork({ latencyFn? })`

```
import { SimNetwork } from '@axona/protocol';

// All edges = 0ms
const net = new SimNetwork();

// 50ms each way
const net = new SimNetwork({ latencyFn: () => 50 });

// Custom per-pair
const net = new SimNetwork({
  latencyFn: (fromId, toId) => Math.abs(hash(fromId) - hash(toId)) %
    100,
});
```

`simTransport({ network, identity, heartbeatMs?, ... })` → `SimTransport`

Factory that builds + wires a `SimTransport` connected to a `SimNetwork`. Same constructor options as `new SimTransport({...})` but with a more concise call site.

```
import { SimNetwork, simTransport } from '@axona/protocol';

const network = new SimNetwork();
const t = simTransport({ network, identity, heartbeatMs: 0 });
await t.start(identity.id);
await t.openConnection(otherIdentity.id);
```

`heartbeatMs: 0` disables heartbeats (recommended for tests; the default heartbeat is helpful for production where peers may go away silently, useless for in-process synchronous flows).

11b. Web transport (browser)

The production browser transport: a WebSocket to the bridge for bootstrap + signaling, and a WebRTC mesh for direct peer-to-peer traffic. It owns the bridge socket lifecycle (reconnect/backoff, ping/pong, stale detection), the version-gate handshake, and the axona/4 authenticated-identity handshake on both the bridge link and every mesh link.

webTransport(opts) → CompositeTransport

```
import { webTransport } from '@axona/protocol/transport/web/index.js';

const transport = webTransport({
  bridgeUrl: 'wss://bridge.axona.net', // required
  identity, // from deriveIdentity() ---
    signs the handshake
  peerVersion: '3.11.0', // your app version (gated by
    the bridge)
  reconnect: true, // auto-reconnect with
    backoff (default)
  // log, autoHandshake, handshakeTimeoutMs, pingIntervalMs,
  // reconnectInitialMs, reconnectMaxMs, WebSocketImpl ...
});
await transport.start(); // resolves after the bridge
  handshake
```

Beyond the Transport contract (send/notify/onRequest/ onNotification/boundPeers/onPeerBound/start/stop), it exposes an observability surface for the UI:

```
transport.bridgeState // 'connecting'|'open'|'stale'|'
  disconnected'|'upgrade-required'
transport.bridgeInfo // { connId, version, kernelVersion, turn
  } | null
transport.bridgeRtt // last ping->pong ms (null until first
  pong)
transport.bridgeRttAvg // mean of the recent RTT window
transport.bridgeNodeId // bridge's 66-hex nodeId (null pre-
  handshake)
transport.onBridgeState((state, detail) => {}); // -> unsub
transport.onWelcome((info) => {}); // -> unsub (replays
  last)
transport.onPingTraffic((dir) => {}); // 'sent' | 'recv'
transport.reconnectNow(); // force an
  immediate reconnect (tab resume)
transport.boundPeers(); // bigint[] ---
  axona/4-authenticated peers
```

Channel binding (finding A-1): each mesh link's axona/4 proof is bound to that connection's DTLS certificate fingerprint, so a fingerprint-rewriting bridge cannot transparently MITM "direct" peer traffic. The bridge is trusted for signaling only, never for the contents of peer-to-peer links. See §17.

12. Handshake

Wire-version handshake helpers used by transports on a fresh signaling channel. Most app code never touches these — they’re plumbed into `simTransport/webTransport/etc.` internally.

```
buildClientHello({ version, nodeId })    // -> frame for client ->
  server
buildServerHello({ version, minVersion }) // -> frame for server ->
  client
parseHello(frame)                       // -> { version, nodeId? }
parseVersion(str)                       // -> { major, minor, patch
  }
compareVersions(a, b)                   // -> -1 | 0 | 1
wireCompatible(peerVersion, minRequired) // -> boolean
performClientHandshake(channel, opts)    // -> handshake result
performServerHandshake(channel, opts)
```

The constants `WIRE_VERSION`, `KERNEL_VERSION`, `UPGRADE_CLOSE_CODE` are exported from the same module — see §16.

Low-level utilities

13. Ed25519 helpers

Web Crypto Ed25519 wrappers. Useful when you want to sign / verify outside of `buildEnvelope` — e.g. for signing direct-message payloads manually.

```
const { pubkey, privateKey } = await generateKeyPair();
// pubkey = Uint8Array(32); privateKey = CryptoKey

const pubHex = await exportPublicKey(pubkey);
const pubKey = await importPublicKey(pubHex);

const sig = await sign(privateKey, dataBytes);    // Uint8Array(64)
const ok = await verify(pubKey, sig, dataBytes);

const signer = makeSigner(privateKey);           // (bytes) =>
  Promise<sig>
const verifier = makeVerifier(pubKey);           // (sig, bytes) =>
  Promise<boolean>
```

Implementation-agnostic: substitute `@noble/ed25519` for runtimes that don’t have Web Crypto Ed25519 (Chrome <110, Safari <17, Firefox <130, Node <20). The `makeSigner` / `makeVerifier` contract is what the rest of the kernel binds against.

14. Hex / ID math

```
ID_BITS           // 264
HASH_BITS         // 256
S2_BITS           // 8
HEX_CHARS         // 66
MAX_ID            // BigInt (2**264 - 1)
MAX_HASH          // BigInt (2**256 - 1)
MAX_S2            // 255

toHex(bigint)      // -> 66-char lowercase hex
fromHex(hex)       // -> BigInt
isHexId(s)         // -> boolean (66-char lowercase hex)
assembleId(s2Prefix, hash) // -> BigInt --- concat 8-bit prefix + 256-bit hash
extractS2Prefix(idBig) // -> number 0..255
extractHash(idBig)  // -> BigInt (bottom 256 bits)
s2PrefixOfHex(hex) // -> number 0..255

xorDistance(a, b)   // -> BigInt
stratumOf(a, b)     // -> 0..264 --- count of leading zero bits
clz264(bigint)      // -> 0..264 --- count of leading zero bits
                    // in a 264-bit value
randomU256()        // -> BigInt --- cryptographically random
```

These are useful when implementing custom routing logic, K-closest overrides, or analytics on top of the address space.

15. Geo utilities

```
geoCellId(lat, lng, bits)           // -> integer --- S2 cell ID at
    the given bit depth
haversine(lat1, lng1, lat2, lng2)   // -> km between two points
roundTripLatency(lat1, lng1, lat2, lng2) // -> ms --- rough fiber
    estimate
randomU32()                          // -> 0..2**32 - 1
```

Sub-path import for the remaining geo helpers (@axona/protocol/utils/geo.js):
getRegion, getContinent, several XOR routing-table builders. Rarely
needed by app code.

16. Constants

```
WIRE_VERSION      // '1.0'      --- wire format major.minor
KERNEL_VERSION    // '2.8.0'    --- kernel semver
AUTH_PROTO        // 'axona/4'  --- authenticated-identity
    handshake tag
UPGRADE_CLOSE_CODE // 4426      --- WebSocket close code for
    version mismatch
```

ID_BITS	// 264
HEX_CHARS	// 66

WIRE_VERSION is what bridges enforce gates against; KERNEL_VERSION is informational and corresponds to the npm release tag.

17. Security model (what the protocol guarantees)

The full, versioned record is the [security changelog](#); this is the developer-facing summary of what's enforced as of kernel v2.11.0. Everything here is **self-authenticating** — no certificate authority, no central trust server, no reputation service.

- ****Authenticated identity (axona/4).** A nodeId's bottom 256 bits are SHA-256(pubkey), and every connection runs a handshake proving the peer holds that key (BIND: pubkey hashes to the id · POSSESS: Ed25519 signature · CHANNEL: the signature is bound to this live connection so a captured proof can't be replayed onto another link). A peer cannot claim an id it doesn't own.
- **Channel binding (untrusted bridge).** Mesh links fold each side's DTLS certificate fingerprint into the signed handshake, so a bridge that terminates DTLS to interpose on a "direct" link produces divergent bindings and the handshake fails. The bridge is trusted for *signaling only*.
- **Pub/sub trust boundary.** A peer can only subscribe *itself* (every path checks the authenticated sender, not a payload-named id) — no reflection/amplification at a victim. A node only hosts topics it's among the K-closest to — no memory-exhaustion via random-topic floods. Publisher signatures are verified at root-axon ingress before fan-out.
- **Verified routing admission (eclipse resistance).** Routing-table entries are *earned*: a peer named in routing gossip is a candidate, admitted only after this node connects and the peer authenticates via axona/4. Forged gossip cannot inject hops. (Observe this via `health().meshDegraded` and `boundCount`.)
- **Key hygiene.** Browser identities can hold a **non-extractable** signing key (`deriveIdentity({ extractable: false })`); `loadIdentity` verifies the private key matches its public key on load.

What the protocol does **not** do: encrypt your application payload (it's opaque bytes — encrypt inside `message` if you need confidentiality), or vouch for what an authenticated peer *says* beyond what

cryptography proves. Open security work is tracked privately; the public changelog lists only resolved items.

A word on stability

`@axona/protocol` v2.x is stable for the application surface documented in §§1–9. The transport + protocol surface (§§10–12, incl. the browser `webTransport` and the `axona/4` security model in §17) is also stable but expect occasional additions (new transport factories, new bootstrap services). Low-level utilities (§§13–16) may grow but won't change incompatibly.

The `AxonaPeer` constructor still accepts a legacy `engine` argument for backward compatibility with pre-Phase-5 callers; new code should pass `domain` instead.

For changelog and migration notes between minor versions, see [<https://github.com/axona-net/axona-protocol/releases>](https://github.com/axona-net/axona-protocol/releases).